ings are more essential to saving lives. Last year, Land Rover partnered with Red Cross for Project Hero, consisting of a Land Rover vehicle with a roof-mounted drone fitted with thermal imaging sensors. The included landing system allows the drone to land on the vehicle while it was in motion. A start-up by the name of Flyability has made significant progress on developing a collision resistant UAV. Also, a team from the Delft University of Technology has been working on an ambulance drone which would be able to deliver defibrillators on demand. The combined capabilities of these drones will help them look for survivors, provide basic aid, locate sources of disasters like forest fires before they can scale up and reach inaccessible places in far shorter times than possible today.

Land Rover Project Hero

FOR PRESERVATION AND HEALTH

Along the same lines as normal deliveries, using drones to deliver medicines to parts of the world where it's not available readily is one of the most anticipated uses of drones. However, the future of drone tech promises bigger things. For instance, using drones to track animals can provide insight into how diseases are spread through nature. Thermal imaging capable drones have been used by the London School of Hygiene and Tropical Medicine in tracking malaria spread via macaque movements in the Philippines, where the disease is a significant problem. Microsoft also has similar tech to capture and test mosquitoes for infectious disease. Drones have also been used to monitor animals for their own health. Projects like the Ocean Alliance use drones like Snotbot to collect samples from whales, alongside drone initiatives that track down poachers.

There's an age-old adage that says we are what we eat. Going by that logic, anything that improves the state of agriculture directly improves human health – which, in this case, would be drones. Autonomous agricultural machinery has been talked about for a while and we are not considering that. What we are talking about is stuff like what Raptor Maps is working on – drones to provide analytics to farmers to help them understand their crop better. With improved image processing capabilities and builds that are smaller and lighter, drones will also be able to handle crops directly for harvest, analysis for disease and even help plant crops, as companies like DroneSeed have established. For research purposes, drones have even been able to pollinate crops, giving rise to speculation that they could someday fill in for the disappearing population of bees.

AUTOMATED INSPECTION

A large chunk of expenses across industries goes into maintenance, and a big part of that is spent on manual inspection. Whether it is infrastructure like pipelines, power lines, telecom networks, insurance or other areas, employees physically visiting sites to check for defects and damages has been the norm for a long time. Autonomous drones are being incorporated by all of the above industries and more to bring down inspection cost and also improve efficiency. For example, Avitas Systems, a GE subsidiary, has already initiated plans for drone inspection for its infrastructure projects.

Snotbot

Airobotics provides solutions that are targeted at mining companies for on-site management and security. These drones are also capable of swapping batteries and cameras once they need changing on their own.

Perhaps the most interesting application of drones for inspection comes from the area of meteorology. Earlier this year, Digital Jersey, conducted a trial to better understand how drones would benefit weather analysis. Currently, most meteorological data is collected by ground stations or high altitude stations. It is quite obvious that using drones for this purpose would provide real-time data from identifiable points in the atmosphere that could even help scientists predict weather better. Along with that, these drones could follow weather changes and patterns as they develop, reducing the chance of unexpected events, like a sudden change in direction for a cyclone, or a flash flood, to a great extent.

OTHER PROJECTS

Essentially, drones are little more than objects that can move from point A to point B on their own, preferably better than humans. It is the technology that modifies them for special use cases that makes their future so promising. We've tried to cover as much as possible within the span of this article, however, there are some futuristic ideas that we kept for last. For example, concepts and experiments around beaming free internet from the skies using drones have been worked on by tech giants like Facebook and Google in the recent past and still hold promise through third-party help. Multiple tech, auto-tech and aviation companies in the world are working on building "drone taxis" to help improve traffic situations and bring down urban travel times significantly. Drones focussed on security are also picking up better tech to provide accurate data and even gain the capability to take action. As the newer regulations in our country kick in during December, it would be interesting to see the direction drones take in our own skies. d